Activity: Group REST API Enhancement and Multi-Client Implementation

Objective

Enhance a REST API and demonstrate how it can be consumed by different client applications across platforms, while practicing teamwork and role distribution.

Background

A demo was presented:

- Server: `service.php`
- Input: `user_id`
- Output: Registration status (`Y/N`)
- Clients:
  - C++ console app
  - Desktop app (PowerBuilder – demonstration only)

This shows that REST APIs are **platform-independent** and can be invoked from different environments.

Group Task Overview

Each group will:

1. **Design and implement an enhanced REST API**
2. **Develop at least two client application**
3. **Document individual contributions**

Group Composition

- Existing groupings

REST API Demo: https://youtu.be/mcU1qV7DLYE

Activity Files

[jcnvdev/school-app: School Application]

Part 1: Server-Side (API Enhancement)

Develop your own REST API with **improved functionality**.

Minimum Requirements

- Accept at least **2 parameters**
- Return **JSON responses**
- Implement **new or expanded logic**, such as:
  - User registration system
  - Login validation
  - CRUD operations (Create, Read, Update, Delete)
  - Multi-status responses (not just Y/N)
  - OR any meaningful system logic

Part 2: Client Application

Create at least **two working client** that consumes your API.

Requirements

- Sends request(s) to your API
- Displays output clearly
- Handles errors properly

Allowed Languages

- C++
- Python
- Java / C#
- JavaScript
- Any programming language

Note: PowerBuilder is **not required**, it was only shown to demonstrate cross-platform capability and different environments.

Part 3: Group Demonstration

Each group must present:

1. API overview (endpoints, parameters, logic)
2. Client application
3. Live demo (request → response)

Part 4: Individual Journal (Required)

Each member must submit a **short journal** (1–2 pages) describing:

Required Contents

- Name and role in the group
- Specific tasks performed (e.g., backend, client, testing)
- Code contributions (describe, not copy full code)
- Challenges encountered
- How they contributed to solving problems

This ensures **individual accountability** within the group.

Submission Requirements

- Source code (server + client)
- API documentation (1–2 pages)
- Individual journals (per member)
- Optional: demo video or screenshots

Grading Rubric (100 points total)

Group Output (70 points)

Criteria	Description	Points
API Functionality	Correct and enhanced logic	25
Client Implementation	Successfully consumes API	20
REST/JSON Usage	Proper structure and design	10
Code Quality	Clean and organized	5
Error Handling	Handles invalid cases	5
Presentation/Demo	Clear and working demo	5

Individual Contribution (30 points)

Criteria	Description	Points
Journal Quality	Clear, complete, reflective	10
Contribution Clarity	Specific and meaningful work	10
Technical Understanding	Shows understanding of system	10

Notes

- Work must be **collaborative**, but journals must be **individual**.
- Equal participation is expected.
- Creativity and additional features will be appreciated.
- Group leader submits the work but the individual journal must be included.